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MIDTERM EXAMINATION
Spring 2009
MTH101- Calculus And Analytical Geometry (Session - 1)

Question No: 1 (Marks: 1) - Please choose one

The set {…,-4,-3,-2,-1,0,1,2,3,4,..} is know as set of ……..

- ► Natural numbers
- ► Integers
- ► Whole numbers
- ► None of these

Question No: 2 (Marks: 1) - Please choose one

The domain of the function $h(x) = \frac{1}{(x-2)(x-4)}$ is

- ► $(-\infty, 2) \cup (2, 4) \cup (4, +\infty)$
- ► $(-\infty, 2\} \cup \{2, 4\} \cup \{4, \infty)$
- ► $(-\infty, 2.5) \cup (2.5, 4.5) \cup (4.5, \infty)$
- ► All of these are incorrect

Question No: 3 (Marks: 1) - Please choose one

$\lim_{x \to 2^+} \frac{1}{x-2} =$

- ► 1
- ► -1
- ► 0
- ► ∞

Question No: 4 (Marks: 1) - Please choose one

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$\frac{2}{3}$

is

- An even number
- None of these
- A natural number
- A complex number

Question No: 5 (Marks: 1) - Please choose one

The set of rational number is a subset of

- Integers
- Natural numbers
- Odd integers
- Real number

Question No: 6 (Marks: 1) - Please choose one

If n-5 is an even integer, what is the next larger consecutive even integer?

- n-7
- n-3
- n-2
- n-4

Question No: 7 (Marks: 1) - Please choose one

If a function satisfies the conditions

f(c) is defined

$\lim_{x \to c^-} f(x)$

Exists

$\lim_{x \to c^-} f(x) = f(c)$

Then the function is said to be

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- ► Continuous at c
- ► Continuous from left at c
- ► Continuous from right at c
- ► None of these

Question No: 8 (Marks: 1) - Please choose one

Tan(x) is continuous every where except at points

- ► $\pm\frac{k\pi}{2}(k=1,3,5,...)$
- ► $\pm\frac{k\pi}{2}(k=2,4,6,...)$
- ► $\pm\frac{k\pi}{2}(k=1,2,3,4,5,6,...)$
- ►
- ► None of these

Question No: 9 (Marks: 1) - Please choose one

A line $x=x_0$ is called ---------- for the graph of a function f if $f(x)\to+\infty \; or \; f(x)\to-\infty$ as x approaches x_0 from the right or from the left

- ► Horizontal asymptotes
- ► None of these
- ► Vertical asymptotes

Question No: 10 (Marks: 1) - Please choose one

---- theorem states that “if f(x)is continuous in a closed interval [a,b] and C is any number between f(a) And f(b) Inclusive ,Then there is at least one number x in the interval [a,b] uch that f(x) =C”

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- ► Value theorem
- ► Intermediate value theorem
- ► Euler's theorem
- ► None of these

Question No: 11 (Marks: 1) - Please choose one

Let L_1 and L_2 be non vertical lines with slopes m_1 and m_2 ,respectively Both the lines are perpendicular if and only if

- ► $m1(-m2) = 1$
- ► $m1m2 = -1$

  $m1 = -\frac{1}{m2}$
- ►
- ► All of these

Question No: 12 (Marks: 1) - Please choose one

The equation of line of the form $y - y_1 = m(x - x_1)$ is known as

- ► Slope intercept form
- ► Point-slope form
- ► Two points form
- ► Intercepts form

Question No: 13 (Marks: 1) - Please choose one

Polynomials are always …… Function

- ► Continuous
- ► Discontinuous

Question No: 14 (Marks: 1) - Please choose one

The solution of the inequality $|x-3| < 3$ is

- ► (-1,7)
- ► (1,7)

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- (1,-7)
- None of these

Question No: 15 (Marks: 1) - Please choose one

The solution set of the inequality $|x+4| \geq 2$ is

- $(-\infty,-6]U[2,+\infty)$
- None of these
- $(-\infty,6]U[-2,+\infty)$
- $(-\infty,-6]U[-2,+\infty)$

Question No: 16 (Marks: 1) - Please choose one

The centre and the radius of the circle $x^2+y^2=a^2$ is

- (1,1),a
- (0,0),1
- None of these
- (0,0) ,a

Question No: 17 (Marks: 1) - Please choose one

The centre and the radius of the circle $(x+5)^2+(y-3)^2=16$ is

- (-5,3) ,4
- (5,-3),16
- (5,-3),4
- None of these

Question No: 18 (Marks: 1) - Please choose one

Consider two function $f(x)=3\sqrt{x} and g(x)=\sqrt{x}$ what is true about these functions

- $f(x).g(x)=3x$

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- $f(x)/g(x) = 3x$
-
- $f(g(x)) = 3x$
- None of these

Question No: 19 (Marks: 1) - Please choose one

Consider two function $f(x) = x^3 and g(x) = (x+9)$ then $fog(x) =$

- $(x+9)^3$
- $x+3$
- $x+9$
- None of these

Question No: 20 (Marks: 1) - Please choose one

The formula $\lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$ is called …….. with respect to x of the function f

- Derivative
- Slope
- Tangent
- None of these

Question No: 21 (Marks: 2)

Find the distance between $A_1(4, 6)$ and $A_2(10, 4)$ using the distance formula.

Question No: 22 (Marks: 3)

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Find solution set for the inequality: $x+3\le 12$

solution

$x+3\le 12$

subtracting 3 from both sides we get

$x+3-3\le 12-3$

$x\le 8$

Question No: 23 (Marks: 5)

Determine whether or not $k(x)=\sqrt{1+\sqrt{2x+1}}$ at $x=2$ is continuous?

Question No: 24 (Marks: 10)

Express the given function in piecewise form without using absolute values

g(x) = |x| + |x-1|
g(x) = |x + x-1|
g(x) = |2x-1|